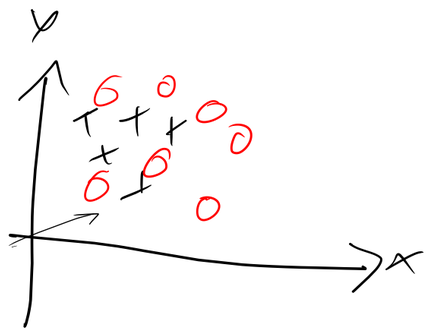
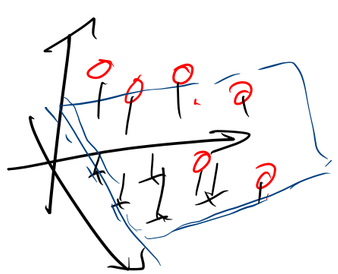
Classification (rep)



note: in the above graph, there are 3 dimensions!

In general, in ML we try to project our features to some higher dimensional space.



Even though a problem may not be linearly separable in some m dimensions, it may be so in some $n > m$ dimensions.

If we can't project (feature engineer interactions etc), then we need other methods (naive bayes, Logistic, ANN)

A general method that works well for large datasets is k-nearest neighbours.

Feature scaling

For most methods we should standard-scale the features.

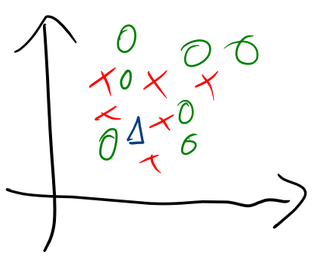
There are some exceptions:

- The response vector is definitely not normal distributed
- The method relies directly on euclidean distances. (eg. knn, kmeans), use min-max

Some notes on Logistic vs kNN:

For large datasets, k-NN usually wins, <u>but</u>

- If classes are intermixed in the data, logistic regression may be the better option
- If there are many distinct classes (no overlap), logistic regression performs poorly compared to k-NN



Logistic

$$\log\left(\frac{P(Y=k \mid X=x)}{P(Y=K \mid X=x)}\right)$$

Find a k that maximizes the probability.

k-NN: statistics ??

kNN has only one hyperparameter: k.

Find optimal k with an elbow plot, run on validation data.

Only min-max scaling for kNN.